

Qualification Pack



Technician - Railway Track Electrification

QP Code: PSS/Q2507

Version: 1.0

NSQF Level: 4

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PSS/Q2507: Technician - Railway Track Electrification

Brief Job Description

The individual in the job will perform various activities related to railway track electrification such as testing of soil, erection of mast, stringing of overhead wires, testing & inspection of the other OHE (Overhead Equipment) etc. in adherence to procedures and standards mentioned in Research Design and Standard Organization (RDSO). The individual also performs erection & commissioning of railway substations.

Personal Attributes

The job requires the individual to have good physical strength and ability to work for long working hours/nights, good eye visibility and ability to communicate. The individual should be ethical and well behaved.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1331: Apply basic health and safety practices for power related work](#)
2. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
3. [PSS/N1336: Working effectively with others](#)
4. [PSS/N2513: Perform erection and commissioning of overhead lines, underground system and equipments](#)
5. [PSS/N2514: Perform erection & commissioning of railway substations](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Operation and Maintenance - Distribution
Country	India
NSQF Level	4
Credits	14



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Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	12th grade Pass (Science Background) OR Completed 2nd year of the 3-year diploma after 10 (regular diploma in relevant filed) OR 10th grade pass (Two year of any combination of NTC/NAC/CITS or equivalent in relevant trade) OR Previous relevant Qualification of NSQF Level (3.5) with 1-2 Years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Relevant Experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-04-PW-00869-2023-V1-PSSC
NQR Version	1.0



Qualification Pack

PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly



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PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1.5
Version	2.0
Next Review Date	31/08/2026



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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	2.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

PSS/N2513: Perform erection and commissioning of overhead lines, underground system and equipments

Description

This unit covers the railway track electrification activities such as testing of soil, erection of mast, stringing of overhead lines, installation of OHE, testing and commissioning of lines, OHE, laying of underground cables etc. as per the RDSO guidelines/ procedures

Scope

The scope covers the following :

- Identify tools and equipment required for erection activity.
- Perform erection of overhead lines, underground system and equipment.

Elements and Performance Criteria

Erect the mast

To be competent, the user/individual on the job must be able to:

- PC1.** organize for PTW (permit to work) in both Power Block and Traffic Block from concerned authorities
- PC2.** perform soil test as per Research Designs and Standards Organisation (RDSO) standards
- PC3.** analyse the data related to the type of civil foundation required based on the soil test results
- PC4.** monitor and inspect the foundation work building process according to RDSO standards
- PC5.** read and interpret the layout plan for erection of mast
- PC6.** select a suitable mast depending on the loading condition, wind load etc.
- PC7.** erect the mast as per the standard procedure laid in RDSO

String lines, underground cables and OHE

To be competent, the user/individual on the job must be able to:

- PC8.** identify various overhead equipment used in railway track electrification Various overhead equipment: Stay tube insulator, tubular stay arm, standard / large bracket, register arm, steady arm, bracket insulator, catenary suspension bracket, register arm hook, drop bracket, anti-wind clamp etc.
- PC9.** mark the anchor points based on the overhead equipment (OHE) used
- PC10.** select a suitable anti-creep point
- PC11.** fix the overhead equipment and carry out stringing of lines
- PC12.** insert droppers as per the required schedule
- PC13.** fix the Auto Tensioning Device (ATD) and adjust the tension of the overhead lines as per the RDSO standards
- PC14.** erect isolators as per the drawing/ layout
- PC15.** erect section insulators on loop lines and cross overs
- PC16.** identify and erect neutral section at suitable location and also differentiate between various types of neutral sections



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- PC17.** select turn out locations and crossing
- PC18.** carry out earthing of every mast and other OHE
- PC19.** provide bonding of masts according to standard norms
- PC20.** obtain necessary clearance from Power Distribution Utilities to increase the pole height or to underground the distribution lines in form of cables to avoid interference of distribution lines and railway lines
- PC21.** perform cable jointing
- PC22.** erect the emergency mast

Test and commission lines, underground cables and OHE

To be competent, the user/individual on the job must be able to:

- PC23.** test all the Tools and Tackles (T and P) for their safe and efficient working
- PC24.** inspect the lines and OHE thoroughly as per the RDSO standards
- PC25.** check the line height suitability for transportation of ODC (Over Dimensional Consignment)
- PC26.** check or measure the stagger implantation as per requirement
- PC27.** check the tension on the wire using a dynamometer
- PC28.** check that the height of the contact wire is as per standards
- PC29.** ensure that the versine angle is in adherence with RDSO standards
- PC30.** ensure that the maximum blow off is under permissible limits
- PC31.** ensure that the electrical clearances are as per the RDSO standards
- PC32.** check that the cable joints are done as per standards
- PC33.** ensure that the cable insulation is as per the manufacturers specifications

Perform energization of overhead lines

To be competent, the user/individual on the job must be able to:

- PC34.** obtain necessary safety clearance regarding overhead line energization from the concerned authority
- PC35.** energize the overhead lines from the respective feeder

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the company
2. relevant health and safety requirements applicable in the work place
3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
4. reporting structure, inter-dependent functions, lines and procedures in the work area
5. principles of electricity
6. principles and practices of electrical safety
7. common electricity terminology and correct interpretation of the same Terminology: e.g. Current, Voltage, Resistance, Inductance, Capacitance, Kilovolt ampere (kVA), Kilowatt (kW), active power, Kilowatt hour (kWh) (unit of electric consumption), Power factor



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8. different types of material and accessories used in railway track electrification Materials and accessories: e.g. mast fitting for various types of insulators such as ST (Suspension type) insulator, stay tube insulator, tubular stay arm, standard / large bracket, register arm, steady arm, bracket insulator, catenary suspension bracket, register arm hook, drop bracket, anti-wind clamp, inclined dropper, suspension clamp, contact wire swivel clip, tubular stay sleeve, tubular stay adjuster, tube cap
9. tools and equipment used in testing, erection and commissioning Tools: e.g. Plier, Screwdriver, Wrench set, Hammer, Drilling machine, Hacksaw / cutting tools, Measuring tape, Pulleys (Force Pulley with sling), Tommy bar, Crimping machine, Round / flat file, Earth rod (discharge rod), torque wrench etc
10. use of soil test results as per RDSO standards
11. use of civil foundation standards as per RDSO
12. applicable limit of tension on overhead line
13. use and function of isolators, interrupters
14. types of Neutral Sections
15. types of turnout and crossing
16. types of bonding used in masts in railway track electrification
17. types and sizes of conductors and cables
18. difference between various types of cables
19. various cable jointing techniques
20. procedure for erection of emergency mas
21. importance of leaving the work area and equipment in a safe and clean condition on completion of the repair and maintenance activities
22. importance of reporting problems in a timely manner
23. personal protective equipment (PPE) and clothing that must be worn during the inspection, repair and maintenance activity and from where can it be obtained PPE: e.g. Safety helmet, safety glove, safety shoe, climbing harness, lanyard and tool belt (when climbing), earth rod (discharge rod), Zola, safety rope
24. how to undertake basic numerical computations and calculations Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
25. how to identify various basic, compound and solid shapes as per dimensions given Basic shapes: square, rectangle, triangle, circle, quadrilaterals Compound shapes: involving squares, rectangles, triangles, circles, semicircles, quadrants of a circle Solid shapes: cube, rectangular prism, cylinder Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity
26. how to use metric systems of measurement
27. how to participate in on-the-job and other learning, training and development interventions and assessments
28. how to maintain current knowledge of application standards, legislation, codes of practice and product/ process developments

Generic Skills (GS)

User/individual on the job needs to know how to:



Qualification Pack

1. fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
2. read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language
3. convey and share technical information clearly using appropriate language
4. clarify task related information with appropriate personnel or technical advise
5. communicate and cooperate with colleagues in the team for better results
6. seek assistance from fellow team members
7. follow organizational rule-based decision-making process
8. plan, prioritize and sequence work operations as per job requirements
9. organize and analyze information relevant to work
10. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
11. work in a team in order to achieve better results
12. identify and clarify work roles within a team
13. avoid and manage distractions to be disciplined at work
14. manage own time for achieving better results
15. identify and express new ideas and initiatives to others
16. participate in improvement procedures including process, quality and internal/ external customer/ supplier relationships
17. exercise restraint while expressing dissent and during conflict situations
18. identify problems with work planning, procedures, output and behavior and their implications
19. prioritize and plan for problem solving
20. communicate problems appropriately to others
21. identify sources of information and support for problem solving
22. modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
23. improve and modify own work practices
24. analyse the problems in the equipment to identify root causes
25. collect information and technical data and define process for equipment testing
26. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
27. develop holistic and comprehensive profile of products based on segregated discrete process stages
28. apply, analyze and evaluate the information gathered from observation,
29. experience, reasoning, or communication, as a guide to thought and action
30. participate in on-the-job and other learning, training and development
31. interventions and assessments
32. clarify task related information with appropriate authority
33. seek to improve and modify own work practices



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34. maintain current knowledge of application standards, legislation, codes of practice and product/process developments

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Erect the mast</i>	8.5	16.5	-	-
PC1. organize for PTW (permit to work) in both Power Block and Traffic Block from concerned authorities	1	3	-	-
PC2. perform soil test as per Research Designs and Standards Organisation (RDSO) standards	2	3	-	-
PC3. analyse the data related to the type of civil foundation required based on the soil test results	1	3	-	-
PC4. monitor and inspect the foundation work building process according to RDSO standards	1	2	-	-
PC5. read and interpret the layout plan for erection of mast	2	2	-	-
PC6. select a suitable mast depending on the loading condition, wind load etc.	0.5	1.5	-	-
PC7. erect the mast as per the standard procedure laid in RDSO	1	2	-	-
<i>String lines, underground cables and OHE</i>	13.5	30.5	-	-
PC8. identify various overhead equipment used in railway track electrification Various overhead equipment: Stay tube insulator, tubular stay arm, standard / large bracket, register arm, steady arm, bracket insulator, catenary suspension bracket, register arm hook, drop bracket, anti-wind clamp etc.	0.5	1.5	-	-
PC9. mark the anchor points based on the overhead equipment (OHE) used	1	1	-	-
PC10. select a suitable anti-creep point	1	2	-	-
PC11. fix the overhead equipment and carry out stringing of lines	1	2	-	-
PC12. insert droppers as per the required schedule	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. fix the Auto Tensioning Device (ATD) and adjust the tension of the overhead lines as per the RDSO standards	1	2	-	-
PC14. erect isolators as per the drawing/ layout	1	2	-	-
PC15. erect section insulators on loop lines and cross overs	1	2	-	-
PC16. identify and erect neutral section at suitable location and also differentiate between various types of neutral sections	1	2	-	-
PC17. select turn out locations and crossing	1	2	-	-
PC18. carry out earthing of every mast and other OHE	1	2	-	-
PC19. provide bonding of masts according to standard norms	1	2	-	-
PC20. obtain necessary clearance from Power Distribution Utilities to increase the pole height or to underground the distribution lines in form of cables to avoid interference of distribution lines and railway lines	1	2	-	-
PC21. perform cable jointing	-	3	-	-
PC22. erect the emergency mast	1	2	-	-
<i>Test and commission lines, underground cables and OHE</i>	7	19	-	-
PC23. test all the Tools and Tackles (T and P) for their safe and efficient working	2	2	-	-
PC24. inspect the lines and OHE thoroughly as per the RDSO standards	1	3	-	-
PC25. check the line height suitability for transportation of ODC (Over Dimensional Consignment)	0.5	1.5	-	-
PC26. check or measure the stagger implantation as per requirement	0.5	1.5	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. check the tension on the wire using a dynamometer	0.5	1.5	-	-
PC28. check that the height of the contact wire is as per standards	0.5	1.5	-	-
PC29. ensure that the versine angle is in adherence with RDSO standards	0.5	1.5	-	-
PC30. ensure that the maximum blow off is under permissible limits	-	2	-	-
PC31. ensure that the electrical clearances are as per the RDSO standards	0.5	1.5	-	-
PC32. check that the cable joints are done as per standards	0.5	1.5	-	-
PC33. ensure that the cable insulation is as per the manufacturers specifications	0.5	1.5	-	-
<i>Perform energization of overhead lines</i>	1	4	-	-
PC34. obtain necessary safety clearance regarding overhead line energization from the concerned authority	1	2	-	-
PC35. energize the overhead lines from the respective feeder	-	2	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2513
NOS Name	Perform erection and commissioning of overhead lines, underground system and equipments
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution, Erection, Installation and Commissioning - Distribution Downstream
NSQF Level	4
Credits	5
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023



Qualification Pack

PSS/N2514: Perform erection & commissioning of railway substations

Description

This unit covers the competencies required by technicians for erection and commissioning of railway substation. This unit also covers the skills and competencies needed for erection and commissioning of other equipment associated with railway substation

Scope

The scope covers the following :

- Demonstrate procedure of erection and commissioning of railway substations.

Elements and Performance Criteria

Perform erection and commissioning of railway substations

To be competent, the user/individual on the job must be able to:

- PC1.** read and interpret layout plan of the substation
- PC2.** ensure that the unloading of the traction transformer at the site is done as per norms
- PC3.** check for any damage to the traction transformer and its components during transportation
- PC4.** assemble the accessories of the traction transformer at the site
- PC5.** carry out pipe and plate earthing to make earth connection and earth mat
- PC6.** ensure that each earth pit be marked with date and next due date for testing of the earth resistance
- PC7.** ensure the availability of double earth connection to each equipment
- PC8.** ensure installation of all substation equipment such as CT (current transformer), CB (circuit breaker), LA (lightening arrestor), PT (potential transformer) etc. is done as per norms
- PC9.** ensure that the voltage level of equipment is as per the specified rating
- PC10.** ensure that the equipment with appropriate voltage level has been fixed appropriately as per drawings
- PC11.** ensure proper connections being made to OHE through feeding post
- PC12.** ensure erection of SP (switching posts) and SSP (Sub-sectioning and Paralleling post) as per standards at appropriate location
- PC13.** erect and install interrupters on SP and SSP
- PC14.** erect the auxiliary transformer for signaling purpose as per specifications and guidelines
- PC15.** check that the painting of all equipment is done as per standards
- PC16.** ensure that the gravel laid in the substation yard is as per specified norms
- PC17.** ensure availability of proper lighting arrangement at the substation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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1. relevant legislation, standards, policies, and procedures followed in the
2. company relevant to own employment and performance conditions
3. relevant health and safety requirements applicable in the work place
4. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
5. escalation matrix and procedures for reporting work and employment related issues
6. principles of electricity Power system: how power flows, generation, transmission and distribution number of bays, number of incoming and outgoing feeders, load management through single or double bus, substation network, ring system, back feed etc
7. line components towers, poles, single circuit, double circuit, overhead, underground conductors and cables
8. function of transformer accessories
9. gantry structure, structure lay out, types of porcelain insulators, overhead conductors, clamps used in station
10. outdoor and indoor equipments like power transformer, circuit breaker, Isolator, LA, CT, PT, and CVT
11. lightening arrestors (LA) functioning
12. difference between various switching stations
13. types of earthing used in grid station, its significance, why earth connection with each equipments, measurement of earth resistance and earthing switch.
14. complete tools, tackles and safety gadgets required in grid station erection and commissioning
15. how to keep records of all equipment like name plate, pre- commission test report and manuals
16. hazards associated with carrying out substation construction and installation process and how they can be minimized
17. how to undertake basic numerical computations and calculations Numerical computations: addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages
18. how to use metric systems of measurement Units and number systems representing degree of accuracy: decimals places, significant figures, fractions as a decimal quantity
19. how to participate in on-the-job and other learning, training and development interventions and assessments
20. how to maintain current knowledge of application standards, legislation, codes of practice and product/ process developments

Generic Skills (GS)

User/individual on the job needs to know how to:

1. fill up appropriate technical forms, process charts, activity logs as per organizational format in English and/or local language
2. read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in English and/or local language
3. convey and share technical information clearly using appropriate language



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4. clarify task related information with appropriate personnel or technical advise
5. communicate and cooperate with colleagues in the team for better results
6. seek assistance from fellow team members
7. follow organizational rule-based decision-making process
8. plan, prioritize and sequence work operations as per job requirements
9. organize and analyze information relevant to work
10. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
11. work in a team in order to achieve better results
12. identify and clarify work roles within a team
13. avoid and manage distractions to be disciplined at work
14. manage own time for achieving better results
15. identify and express new ideas and initiatives to others
16. participate in improvement procedures including process, quality and internal/ external customer/ supplier relationships
17. exercise restraint while expressing dissent and during conflict situations
18. identify problems with work planning, procedures, output and behavior and their implications
19. prioritize and plan for problem solving
20. communicate problems appropriately to others
21. identify sources of information and support for problem solving
22. modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
23. improve and modify own work practices
24. analyse the problems in the equipment to identify root causes
25. collect information and technical data and define process for equipment testing
26. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
27. develop holistic and comprehensive profile of products based on segregated discrete process stages
28. apply, analyze and evaluate the information gathered from observation,
29. experience, reasoning, or communication, as a guide to thought and action
30. participate in on-the-job and other learning, training and development
31. interventions and assessments
32. clarify task related information with appropriate authority
33. seek to improve and modify own work practices
34. maintain current knowledge of application standards, legislation, codes of practice and product/process developments

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform erection and commissioning of railway substations</i>	30	70	-	-
PC1. read and interpret layout plan of the substation	2	4	-	-
PC2. ensure that the unloading of the traction transformer at the site is done as per norms	1	3	-	-
PC3. check for any damage to the traction transformer and its components during transportation	1	3	-	-
PC4. assemble the accessories of the traction transformer at the site	2	6	-	-
PC5. carry out pipe and plate earthing to make earth connection and earth mat	2	6	-	-
PC6. ensure that each earth pit be marked with date and next due date for testing of the earth resistance	2	4	-	-
PC7. ensure the availability of double earth connection to each equipment	2	3	-	-
PC8. ensure installation of all substation equipment such as CT (current transformer), CB (circuit breaker), LA (lightening arrestor), PT (potential transformer) etc. is done as per norms	2	4	-	-
PC9. ensure that the voltage level of equipment is as per the specified rating	2	3	-	-
PC10. ensure that the equipment with appropriate voltage level has been fixed appropriately as per drawings	2	4	-	-
PC11. ensure proper connections being made to OHE through feeding post	2	4	-	-
PC12. ensure erection of SP (switching posts) and SSP (Sub-sectioning and Paralleling post) as per standards at appropriate location	2	4	-	-
PC13. erect and install interrupters on SP and SSP	3	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. erect the auxiliary transformer for signaling purpose as per specifications and guidelines	2	5	-	-
PC15. check that the painting of all equipment is done as per standards	1	4	-	-
PC16. ensure that the gravel laid in the substation yard is as per specified norms	1	4	-	-
PC17. ensure availability of proper lighting arrangement at the substation	1	4	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2514
NOS Name	Perform erection & commissioning of railway substations
Sector	Power
Sub-Sector	Distribution
Occupation	Erection, Installation and Commissioning - Distribution, Erection, Installation and Commissioning - Distribution Downstream
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70% aggregate for the QP.



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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	39	61	0	0	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
PSS/N1336.Working effectively with others	34	56	0	0	90	10
PSS/N2513.Perform erection and commissioning of overhead lines, underground system and equipments	30	70	-	-	100	35
PSS/N2514.Perform erection & commissioning of railway substations	30	70	-	-	100	35
Total	153	287	0	0	440	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
RDSO	Research Design And Standard Organization
OHE	Over Head Equipment
ATD	Auto Tensioning Device
PTFE	Poly Tetra Fluoro Ethene
ODC	Over Dimensional Consignment
EIG	Electrical Inspector General
CRS	Commission Of Railway Safety
TSS	Traction Sub-Station
SP	Sectioning And Paralleling Post
SSP	Sub-Sectioning And Paralleling Post
YS	Yard Supply Post
MH	Material Handling
CT	Current Transformer
PT	Potential Transformer
CVT	Capacitive Voltage Transformer
LA	Lightening Arrestor
CB	Circuit Breaker

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.